

Roll No.

Total No. of Pages : 02

Total No. of Questions : 16

B.Sc. (IT) (2015 to 2018) / BCA (2014 to 2018) (Sem.-1)

**PROGRAMMING IN C**

Subject Code : BSIT/BSBC-102

M.Code : 10044

Time : 3 Hrs.

Max. Marks : 60

**INSTRUCTIONS TO CANDIDATES :**

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and students have to attempt any FOUR questions.

**SECTION-A**

**Answer briefly :**

1. List the different keywords in C.
2. How is a program compiled and executed?
3. What is the role of Header files in C program?
4. Differentiate between constants and variables.
5. What are the various Input/output statements in C?
6. List the various operators used in C.
7. How are pointers declared and initialized?
8. Differentiate between Call by value and Call by reference.
9. What is the use of storage classes?
10. Write the syntax for do-while loop.

## SECTION-B

11. a) What is the use of Algorithms in developing a program? How is it different from Flowcharts?  
b) Write a C program to reverse a string.
12. Describe the following :  
a) Unions and Structures  
b) Macros in C
13. What do you mean by Recursion? What is the use of recursion in C programs? Write a program to find factorial of a number using recursion.
14. a) Describe the use of Arrays in C program with an example.  
b) List the various operators used in C along with its precedence.
15. Write brief notes on :  
a) Difference between If-else and Nested- if statement.  
b) Use of Switch statement in a C program.
16. a) What are Functions? How are Functions declared and parameters passed to functions?  
b) Write a C program to print the average of all numbers from 1 to 50.

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**Total No. of Pages: 02**

**Total No. of Questions: 07**

**BCA / BSc. (IT) / DCA (Sem. – 1)**

# PROGRAMMING IN C

**Subject Code: BSBC-102/BSIT-102**

**M Code: 10044**

**Date of Examination : 12-01-2023**

**Time: 3 Hrs.**

**Max. Marks: 60**

**INSTRUCTIONS TO CANDIDATES:**

1. **SECTION-A** is **COMPULSORY** consisting of **TEN** questions carrying **TWO** marks each.
2. **SECTION-B** contains **SIX** questions carrying **TEN** marks each and students have to attempt any **FOUR** questions.

## SECTION-A

1. Answer the following:
  - a) What are global and local variables?
  - b) What is the use of Flowcharts?
  - c) What are the advantages of Recursion?
  - d) Differentiate between Continue and Goto statements.
  - e) What do you mean by Structures in C?
  - f) List the various data types used in C.
  - g) Briefly describe the use of Pointers in C.
  - h) Differentiate between Call by value and Call by reference.
  - i) What is the use of Macros?
  - j) Write the syntax for function declaration in C.

## SECTION-B

2. a) What are the different steps in developing a program? What is the role of debugging?  
b) With the help of a program, explain the handling of Strings in C programs.
3. Describe the following:  
a) Unions  
b) File handling in C
4. What do you mean by Recursion? What is the use of recursion in C programs? Write a program to find the factorial of a number using recursion.
5. a) Describe the structure of a C program with an example.  
b) List the various operators used in C along with their precedence.
6. Write brief notes on:  
a) Difference between While and Do-while loop.  
b) Use of Jump statements in a C program.
7. a) What are Arrays? How are arrays declared and initialized? What is the difference between single and multi-dimensional arrays?  
b) Write a C program to print all the even numbers between 1 and 100.

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**BCA / BSc. (IT) / DCA (Sem. – 1)**

# PROGRAMMING IN C

**Subject Code: BSBC-102/BSIT-102**

**M Code: 10044**

**Date of Examination : 12-01-2023**

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## SECTION-A

1. Answer the following:
  - a) What are global and local variables?
  - b) What is the use of Flowcharts?
  - c) What are the advantages of Recursion?
  - d) Differentiate between Continue and Goto statements.
  - e) What do you mean by Structures in C?
  - f) List the various data types used in C.
  - g) Briefly describe the use of Pointers in C.
  - h) Differentiate between Call by value and Call by reference.
  - i) What is the use of Macros?
  - j) Write the syntax for function declaration in C.

## SECTION-B

2. a) What are the different steps in developing a program? What is the role of debugging?  
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b) Write a C program to print all the even numbers between 1 and 100.

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**BCA (2019 Batch) (Sem.-1)**  
**PROBLEM SOLVING USING C**

**Subject Code : UGCA-1903**

**M.Code : 76963**

**Date of Examination : 14-01-2023**

**Time : 3 Hrs.**

**Max. Marks : 60**

### INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains SIX questions carrying TEN marks each and students have to attempt any FOUR questions.**

## SECTION-A

1. **Write briefly :**
  - a) What do you mean by function prototype? How it is defined?
  - b) Compare local and global variables with example.
  - c) Explain the conditional operator with example.
  - d) Discuss structure of a C program.
  - e) What are header files? Why these are required? Give example.
  - f) What is the relationship between formal arguments and actual arguments?
  - g) What is a pointer? How a pointer variable is declared, initialized and accessed?
  - h) What are identifiers and keywords? Give example.
  - i) What is a constant? How it is declared and used?
  - j) Define Array? How 1-D and 2-D arrays are declared? Give example.

## SECTION-B

2.
  - a) Explain various data types of C with example. Give memory requirement of each.
  - b) Define recursion. Write a recursive function to compute factorial of a given number.
3.
  - a) Explain if-else, nested-if and switch statements of C with examples.
  - b) Compare call by value and call by reference parameter passing with examples.
4.
  - a) Draw a flow chart to print largest of 3 numbers.
  - b) What are constants and variables? Discuss the rules how constants and variables are defined. Give examples.
5.
  - a) Define string. Explain various operations which can be applied on strings with examples.
  - b) What are two-dimensional arrays? How these are declared? Write a program to add two matrices.
6.
  - a) What are pointers arrays? How you can declare and access 1-D and 2-D array elements using pointers? Discuss with examples.
  - b) Write a program using pointers to find largest element of the array.
7.
  - a) Write a program to design a structure for 30 students of a class with Roll No, Name and Marks. Find the average marks of the class.
  - b) Explain various file operations available in C with examples.

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